

| Final Lab Report: Magnetoelastic Generators for Energy and Biomonitoring |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
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| Bioengr 167: Bioengineering Laboratory 2024 Winter                       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| Group Members                                                            | Ira Kulkarni, Jeannette Merlos, Anahy Barajas Bautista, Allen Kuo, Lisbeth Sanchez, Alice Matsuda, Zaid Slayyeh                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| Time and Location                                                        | Sections: 1A, 1B, 1C Boelter Hall 7736                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
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| Abstract                                                                 | <p>We developed a wearable magnetoelastic generator (MEG) embedded in a compression shirt for real-time respiratory monitoring. The device utilizes a mix of Ecoflex and magnetic powder, with initial testing exploring various copper coil configurations to optimize current output and signal stability. Reducing the Ecoflex to magnetic powder ratio and increasing coil pattern diameter significantly enhanced sensitivity to a broad range of breathing patterns, demonstrating the device's versatility across different users. Integration with an Arduino circuit facilitated data transmission directly to a laptop, highlighting the system's portability and potential for real-time monitoring applications. Despite challenges in Bluetooth connectivity, the final MEG design shows promise for health monitoring in athletes and the general population, offering a non-toxic, biocompatible, and durable alternative to traditional battery-powered systems.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
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| Introduction                                                             | <p>Please tell the readers why your study is important to the research community from three pathway:</p> <ol style="list-style-type: none"> <li>1. What are the current limitations in the community?</li> <li>2. Why would your study help address these concerns and advance the current field?</li> <li>3. A brief statement to describe your experimental achievements. This part could be very much like the Abstract section, but with more details.</li> </ol> <p>Respiration is one of the main vital signs measured when making a medical diagnosis. By tracking an individual's respirations, certain disorders such as asthma, COPD, and respiratory infections can be easily diagnosed. On top of its medical diagnosing abilities, the ability to track respirations can also be used for commercial purposes. With a biomedical device that can track respirations, athletes can acquire more information on their condition and physical wellness. Aside from athletics, tracking respirations can be used for stress &amp; anxiety management as well. If a mobile app was able to receive information on an individual's respirations, it would be able to offer breathing techniques to reduce anxiety while evaluating if they're being done correctly. With this information, it was determined that a wearable bioelectronic that could track respirations would have wide medical and commercial applications.</p> <p>To create a respiration tracker, the applications of MEGs were explored and incorporated with a compression shirt. Magnetoelastic generators (MEGs) utilize two properties of magnets to generate electrical energy from mechanical energy. These two properties are the magnetoelastic effect and electromagnetic induction. The magnetoelastic effect is the concept that magnetic objects experience a change in magnetic properties when they undergo mechanical stress. Electromagnetic induction is the production of current as a result of changing magnetic fields through a solenoid. Combining these two concepts creates the two layers that are present in MEGs. The magnetomechanical coupling layer utilizes the magnetoelastic effect, converting mechanical change into magnetic change, and the magnetic induction layer, converting magnetic change to electricity.</p> |

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|                   | <p>By attaching an MEG into a compression shirt, it produces current whenever an individual breathes, allowing for the tracking of respirations. Through testing, it was evaluated that the MEG can accurately detect different types of breathing. Currently in the wearable electronics field, devices that detect respirations all exist as belts and may not be as comfortable as the shirt that has been designed. This seems to be the main limitation currently with respiration trackers. By utilizing a shirt design, the device becomes a lot more wearable on a daily basis allowing for respiration tracking of not only athletes but anyone that wants to gather information on how they're breathing.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| <b>Procedures</b> | <p><b>Fabricate a magnetoelastic generator 1</b></p> <ol style="list-style-type: none"> <li>1. Add a 1:1 ratio of Part A and Part B till you get 10.70g of Echoflex to a MEG mold</li> <li>2. Add 10.80g of Magnetic Powder to the Echoflex</li> <li>3. Mix till smooth</li> <li>4. Let dry for 4 hours</li> <li>5. Cut into a two 3- 5cm circles</li> <li>6. Magnetize</li> <li>7. Sew both together using cotton thread</li> </ol> <p><b>Fabricate a magnetoelastic generator 2</b></p> <ol style="list-style-type: none"> <li>1. Add a 1:1 ratio of Part A and Part B till you get Echoflex of 20.22g to a MEG mold</li> <li>2. 40.270g of Magnetic Powder to the Echoflex</li> <li>3. Mix till smooth</li> <li>4. Let dry for 4 hours</li> <li>5. Cut into a two 3- 5cm circles</li> <li>6. Magnetize</li> <li>7. Sew both together using cotton thread</li> </ol> <p><b>Fabricate a Conductive coil</b></p> <ol style="list-style-type: none"> <li>1. Turns some Copper Wire 100 times to make it a 3-5cm circle</li> <li>2. Turns some Copper Wire 75 times to make it a 3-5cm circle</li> <li>3. Turns some Copper Wire 100 times to make it a 3-5cm circle</li> <li>4. Strip both ends of each copper wire</li> </ol> <p><b>Fabricate a Compression Shirt Design</b></p> <ol style="list-style-type: none"> <li>1. Cut a L shape design from compression shirt 2</li> <li>2. Sew onto the diaphragm of compression shirt 1 and leaving one side unsewn to create a pocket</li> <li>3. Cut a long strip from compression shirt 2 and sew a tunnel on to compression shirt 1 along the seam</li> <li>4. Slide copper wire through the tunnel and place between the MEG</li> <li>5. Place MEG in the pocket</li> <li>6. Connect the ends of the copper wire to the Current Amplifier Device (Model SR570 Low Noise Current Preamplifier)</li> </ol> <p><b>TESTING</b></p> <p>To begin testing it is important for the compression shirt to fit snug on the person in order to gain accurate results. The shirt can not be loose fitting as this will not accurately pick up the breathing pattern of the person wearing it. Additionally it is</p> |

important to also ensure that the MEG is placed in the center of the diaphragm to get the strongest breathing pattern possible. Not centering the MEG or having it somewhere other than the diaphragm will not give the best results.

1. Put on fabricated shirt and make sure the MEG is centered on the diaphragm
2. Using the wires from the shirt connect them to the Current Amplifier Device in the lab

#### **Normal Breathing**

1. Have the person wearing the shirt breath normally
2. Record data

#### **Rapid Breathing**

1. Have the person wearing the shirt breath quickly
2. Record data

#### **Deep Breathing**

1. Have the person wearing the shirt take deep breaths
2. Record Data

#### **Arduino Testing**

1. Open Arduino on your computer
2. In void setup write the following code

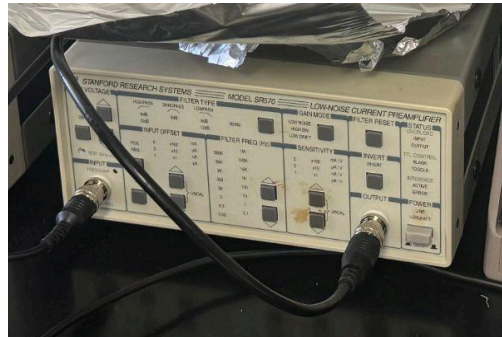
```
pinMode(A0, INPUT);  
Serial.begin(9600);
```

3. In void loop write the following code

```
int val = analogRead(A0);  
Serial.println(val);
```

4. Plug in the circuit board containing an amplifier and connect the wires of the shirt to the input wires on your board.
5. Follow the testing protocols for normal, rapid, and deep breathing.

### Instruments to Test MEG Device



**Figure 1a. Current Amplifier Device  
(Model SR570 Low Noise Current Preamplifier)**

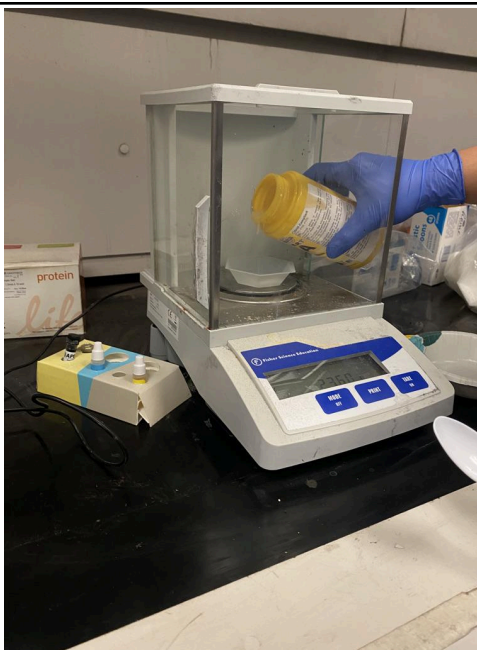
Note: We collected current output for all of our testing and trials using the device in the image above. The SR570 stands as a low-noise current preamplifier capable of current gains as large as 1 pA/V. Its combination of high gain, wide bandwidth, minimal noise, and user-friendly features make the device ideal for a variety of measurements.



**Figure 1b. Arduino Software IDE (w/Breadboard)**  
The Arduino Breadboard which is connected to the laptop with an adaptor allowed us to display the signal outputs on a personal device.

### MEG Fabrication

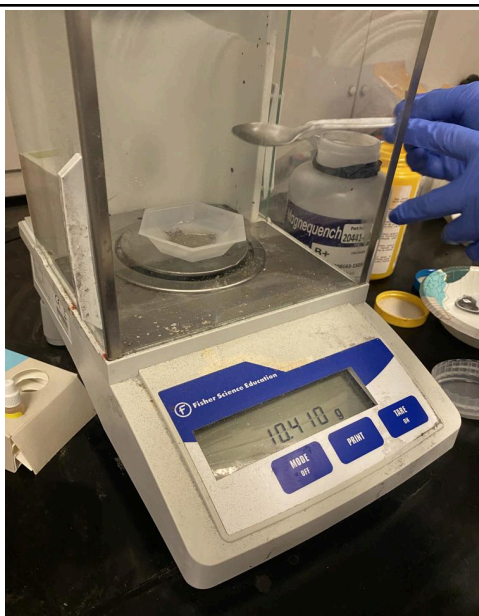
The following images will depict the **general fabrication** of the MEG, including the materials and lab tools used to create the casting of the MEG:



**Figure 2a. Weighing 5g of Part A (Ecoflex)**  
Ecoflex components require 1:1 Ratio

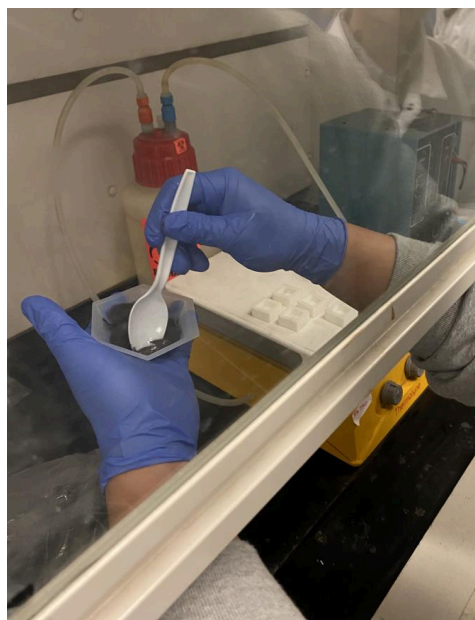


**Figure 2b. Weighing 5g of Part B (Ecoflex)**  
Ecoflex components require a 1:1 Ratio;  
therefore, 5g of Part A and 5g of Part B

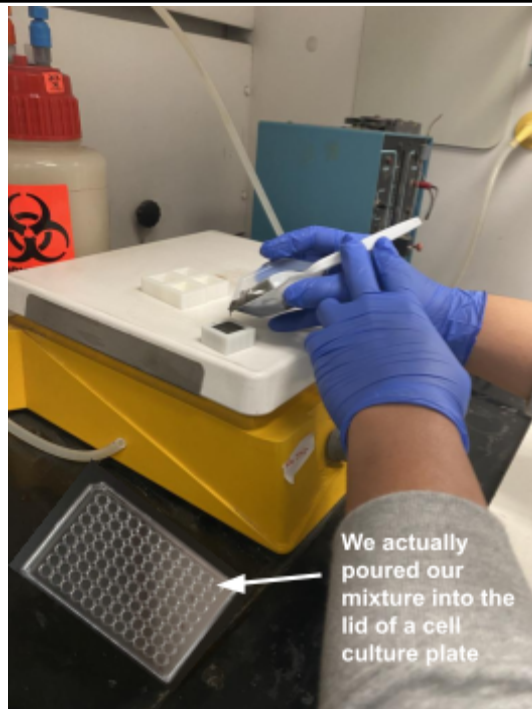


**Figure 2c. Weighing 10g of Magnetic Powder (Magnequench)**

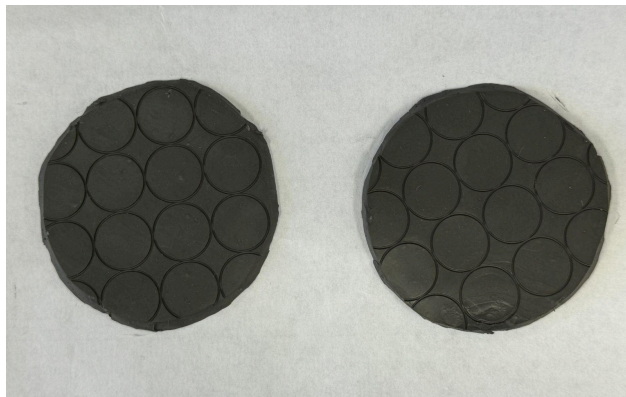
Note: This powder in its original state is non-magnetized.



**Figure 2d. Mixing Ecoflex Parts A/B & Magnetic Powder**



**Figure 2e. Pouring Mixture of Ecoflex & Magnetic Powder into MEG Mold**  
The mixture cured for about two days in the fume hood, and then we cut out two circles that were 5 mm in diameter so that TA Joyce could magnetize the MEG parts



**Figure 2f. Exterior View of MEG Parts**

Note: Both circular parts of MEG were 5x5 mm in diameter

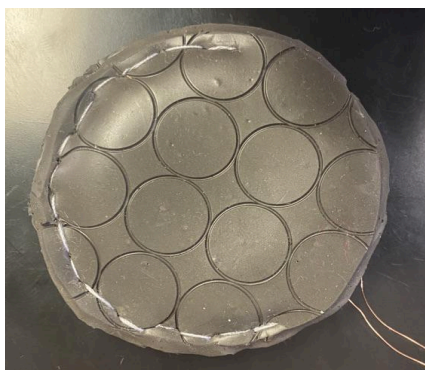


**Figure 2g. Interior View of MEG Parts**



**Figure 2h. MEG Parts Sewn Together**

MEG "Sandwich" (1.0) in which we sewed the MEG parts about halfway to leave space for inserting the coiled copper wire



**Figure 2i. MEG Parts Sewn Together**

MEG "Sandwich" (2.0) was produced by following very similar fabrication steps to our first MEG with the only modification being a 1:2 Overall Ratio of Ecoflex to Magnetic Powder, specifically 20.22 g of Ecoflex and 40.270 g of Magnetic Powder.

### **Integration of MEG in Compression Shirt**



**Figure 3. MEG “Sandwich” Device in Compression Shirt**

We placed our MEG into a pocket on the inside of the shirt. The coils attached to the MEG device were then sewn along the seams of the shirt as depicted in the image above.

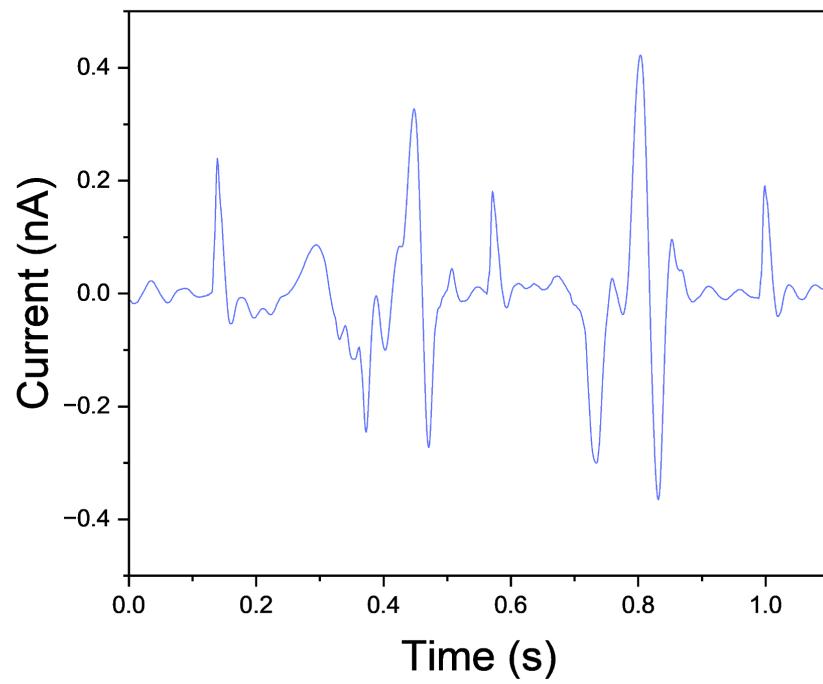
### **MEG Testing**

In the following analysis of our results, we will include the following major findings:

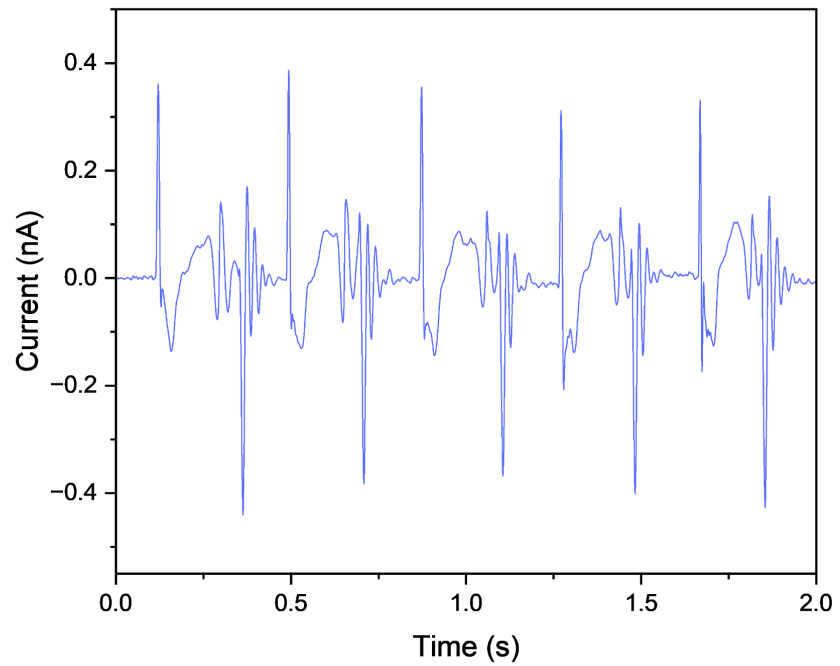
- Initial testing of the individual MEGs we fabricated in the lab
- Formal testing of the integrated MEG Device (placed in compression shirt)
- Formal testing with modifications to the MEG Device and the reasoning for such improvements
- Continued testing & investigation with a newly fabricated MEG, as well as the reasoning for additional modifications to the MEG Device
- Advancements to our Compression Shirt MEG device using Arduino, specifically in monitoring respiration and collecting signal outputs

### **Test Day #1 (Individual MEG Testing)**

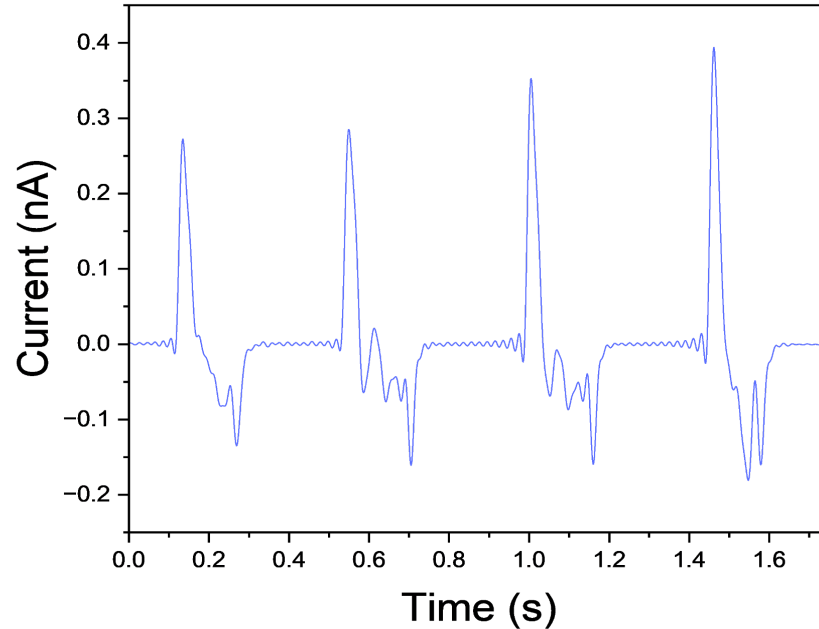
On the first day of our testing, we started by conducting test trials on the individual MEG we had fabricated. We inserted **three variations of coiled copper wire (25, 50, and 100 turns of copper wire)** and compared their current output when placed inside of our MEG “Sandwich” Device. Our data was filtered using the Origin Pro Software and displayed in the following graphs.



**Graph 1. 25 smaller coil turns - Current Output  
(Test Trial of MEG “Sandwich”)**



**Graph 2. 50 smaller coil turns - Current Output  
(Test Trial of MEG “Sanwich”)**



**Graph 3. 100 smaller coil turns - Current Output  
(Test Trial of MEG “Sandwich”)**

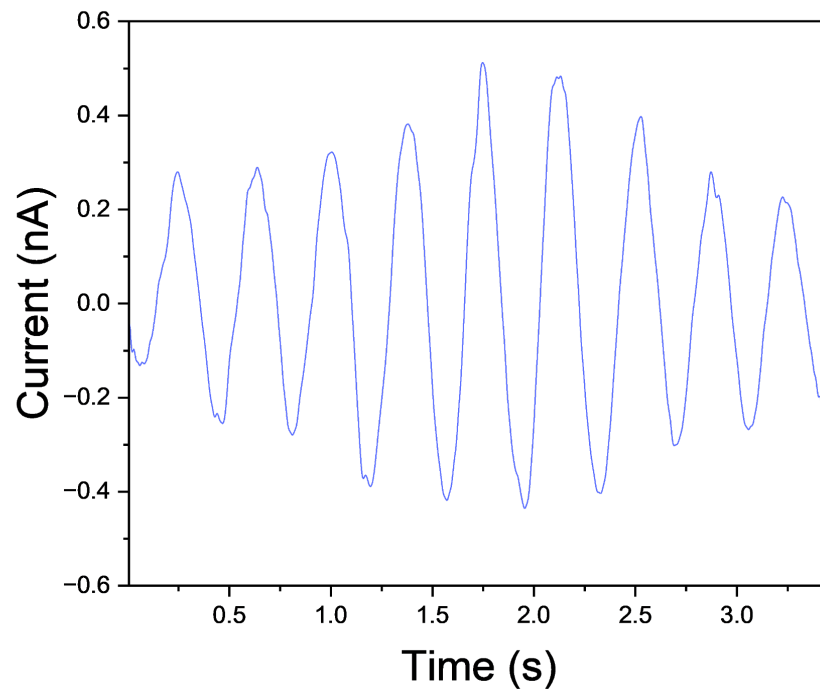
#### **Test Day #1 (Integrated MEG Testing on Zaid)**

After the initial testing of the MEG Device, we noted that the 50 and 100 coil turns of copper wire had the strongest and most consistent current output. We chose to do our testing with these two variations of coiled copper wire for the formal testing on our team member, Zaid, who would be wearing the compression shirt.



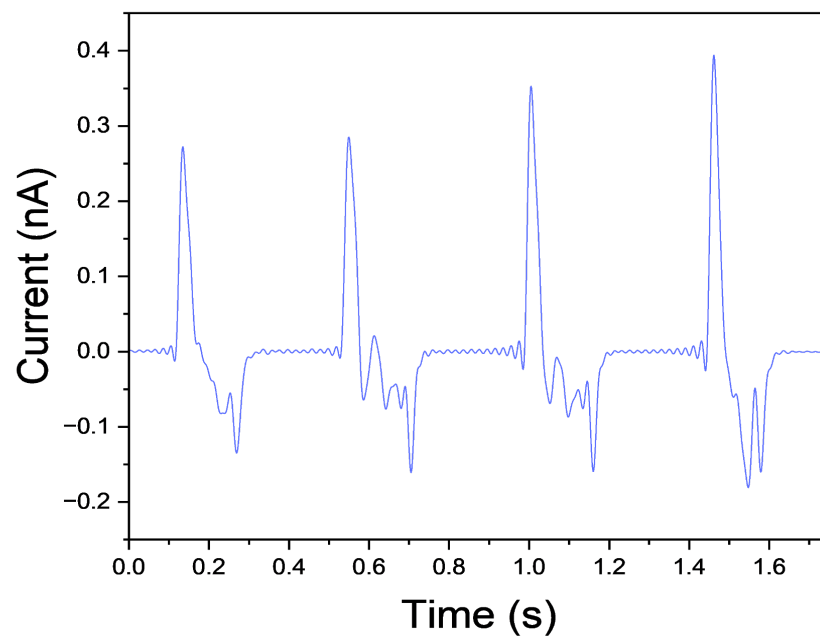
**Figure 4. Monitoring Respiration of Zaid**

As seen in the image, Zaid is wearing the Compression Shirt with the integrated MEG Device. The copper wires that come out from the bottom of the shirt were then connected to the testing device to obtain signal outputs.



**Graph 4. 50 smaller coil turns - Current**  
**Zaid's respiration after running + breathing hard)**

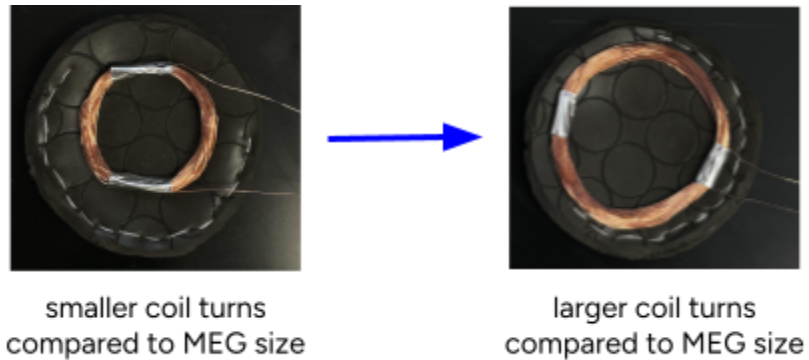
We had Zaid run around the 7th floor of Boelter Hall to increase his breathing rate as this allowed us to have the strong current outputs displayed in the graph above/below.



**Graph 4. 100 smaller coil turns - Current**  
**Zaid's respiration after running + breathing hard)**

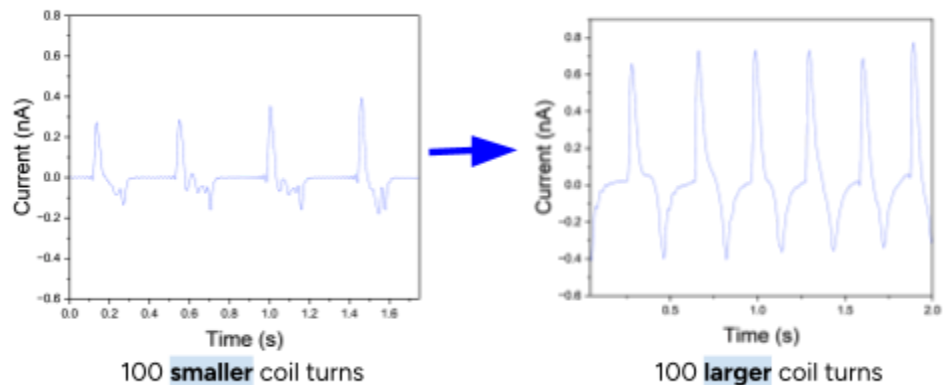
## Test Day #2

After the first testing day, we determined that we did not have as strong of a current output as we had hoped to have. We wanted to find a way to improve our output and asked TA Joyce during lab section if she had any suggestions or improvements for obtaining stronger signals. She suggested that we make larger coil turns that are closer to the size of the MEG, so we followed her advice, made the modifications to our copper wire, and had the following results.



**Figure 5. Modifications to Coiled Copper Wire**

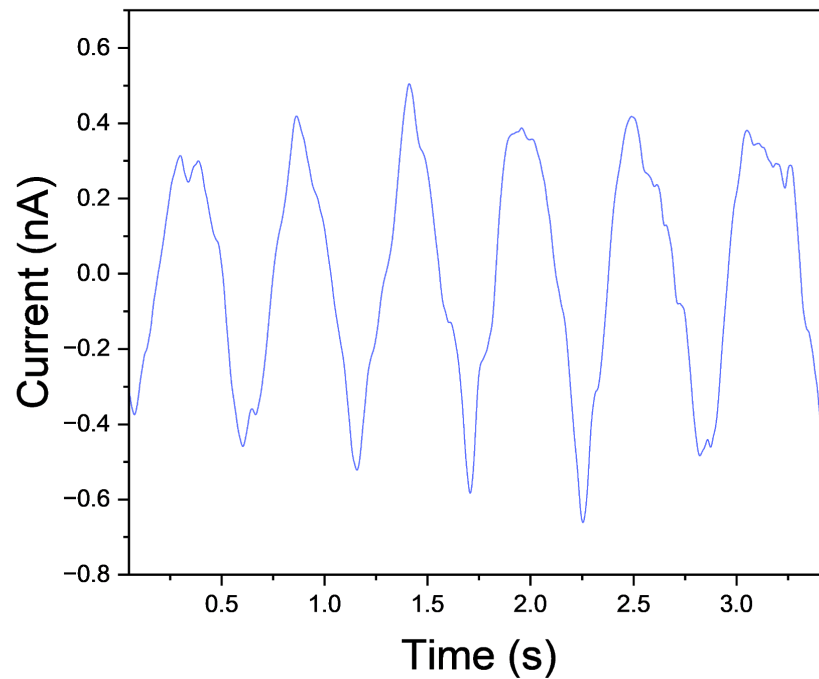
## Test Day #2 (Individual Testing of Modified MEG)



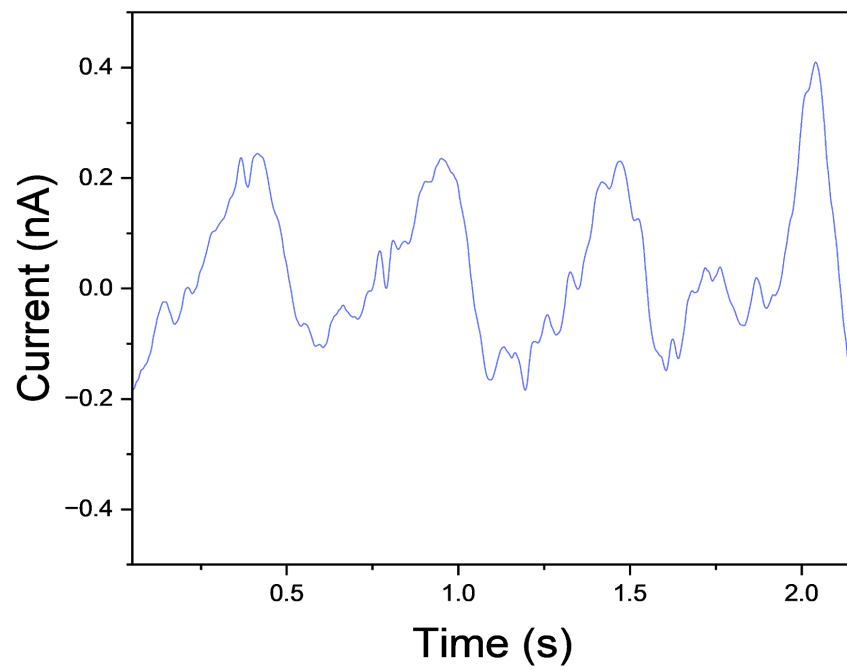
**Graphs 5-6. Comparison of Current Output**

### **Test Day #2 (Integrated MEG Testing on Group Members)**

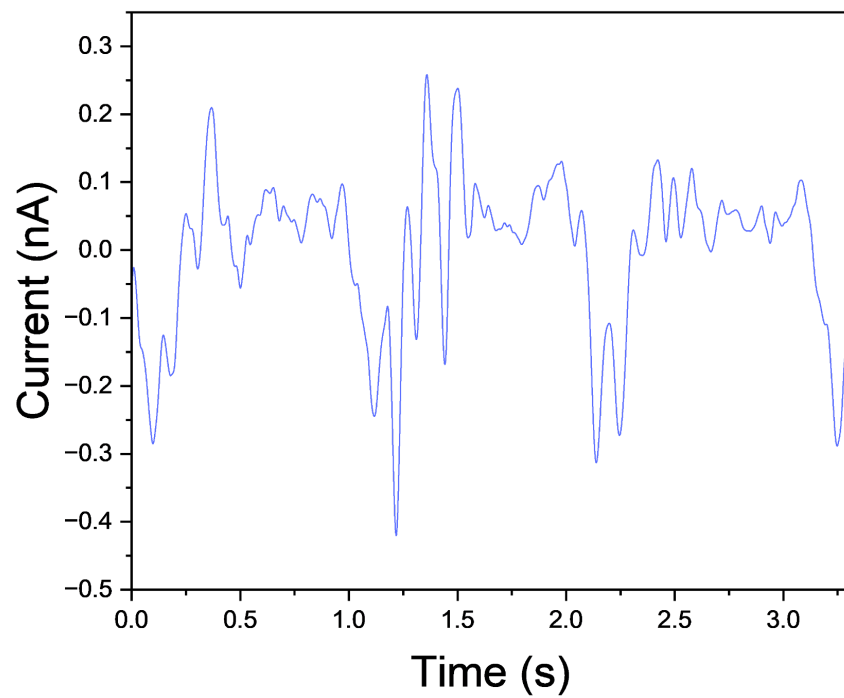
We then decided to test on 4 other groupmates (Allen, Lisbeth, Anahy, and Jeannette) to see if we could get stronger signal outputs for respiration monitoring. In the following graphs we can note the differences in the breathing patterns of our group members.



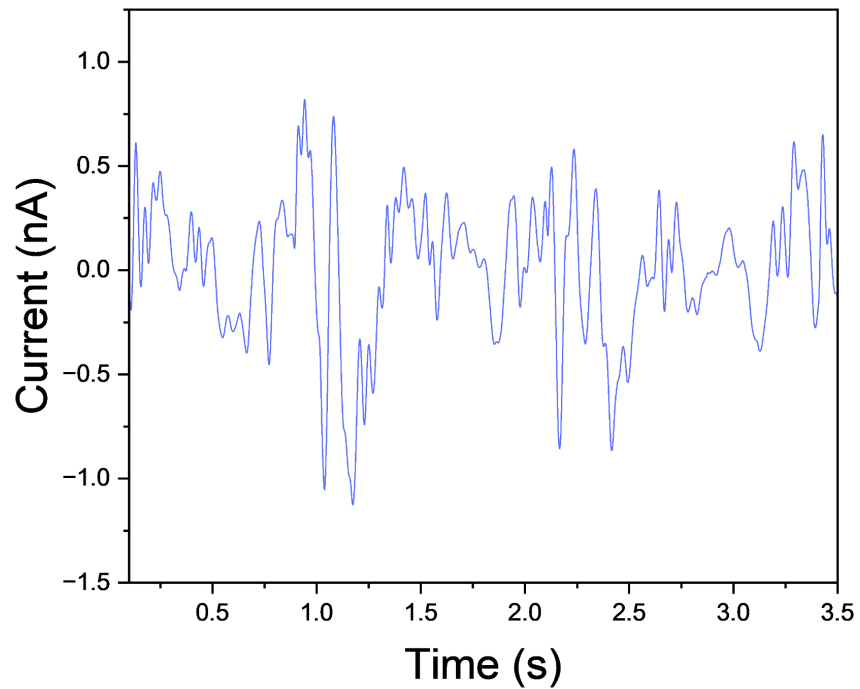
**Graph 7. 100 larger coil turns - Current  
(Allen breathing deeply while standing)**



**Graph 8. 100 larger coil turns - Current  
(Lisbeth breathing while standing)**



**Graph 9. 100 larger coil turns - Current  
(Anahy breathing deeply while standing)**



**Graph 10. 100 larger coil turns - Current  
(Jeannette breathing deeply while standing)**

### **Test Day #3**

Using **larger coil turns and a new MEG** for our testing we monitored respiratory patterns of only one of our group members, Zaid. The modifications to our MEG and analysis of his results are seen below:

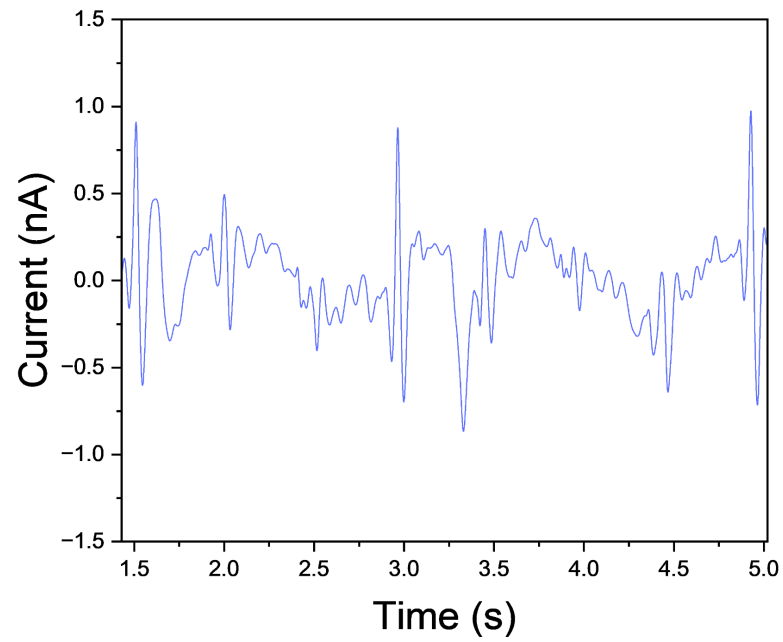


**MEG 1.0**  
1:1 Ratio of  
Ecoflex to  
Magnetic Powder

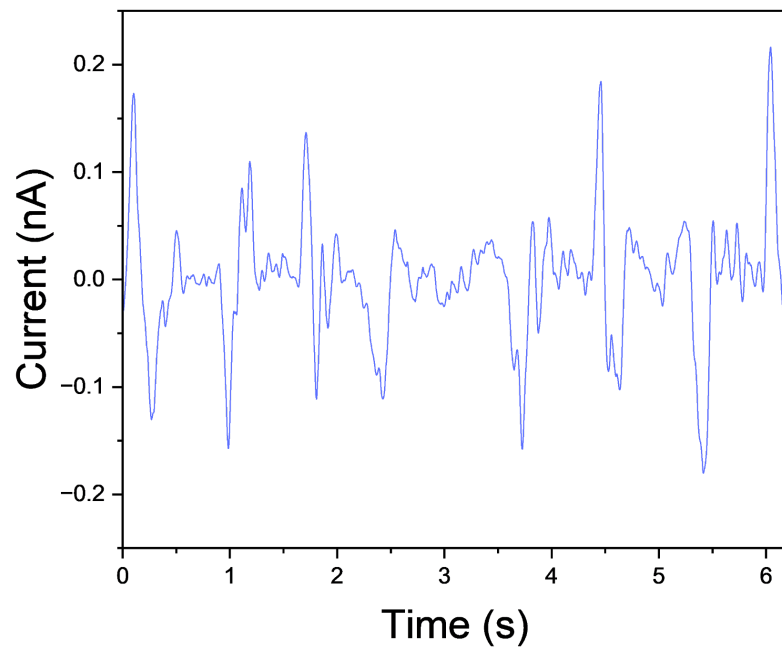


**MEG 2.0**  
1:2 Ratio of Ecoflex  
to Magnetic  
Powder

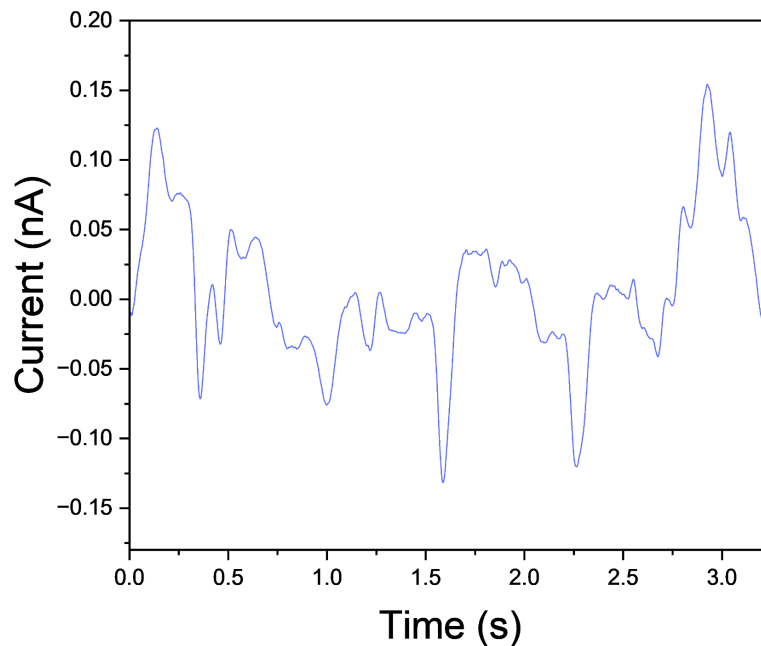
**Figure 6. Comparison between MEG 1.0 and MEG 2.0**



**Graph 10. 100 larger coil turns - Current  
Zaid's breathing patterns directly after running**



**Graph 11. 100 larger coil turns - Current  
Zaid's breathing deeply after a few minutes**

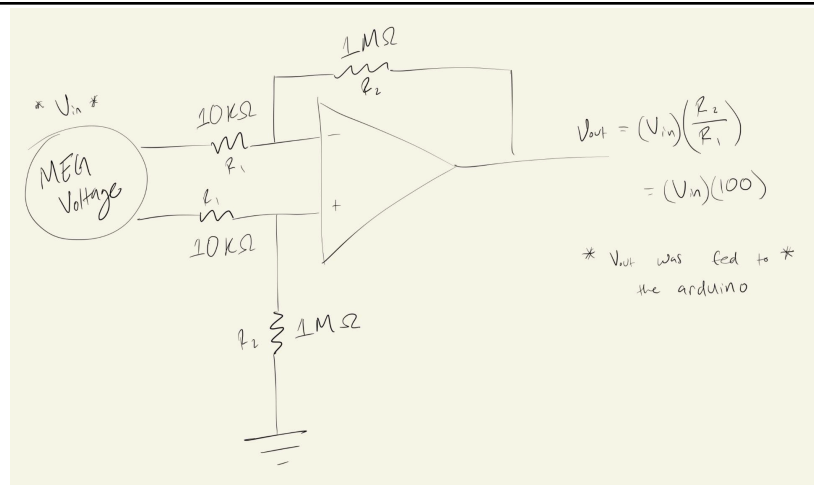


**Graph 12. 100 larger coil turns - Current**  
**Zaid's normal breathing after resting**

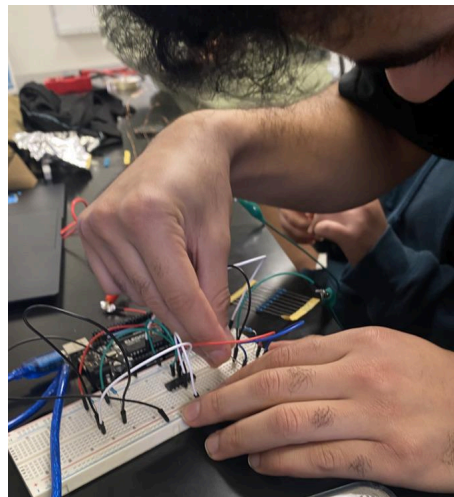
### **Test Day #3 (Integration of Arduino)**

During our final testing day, we also incorporated the Arduino components in our formal testing. We had been working through various lab sections with TA Xiao to connect our MEG device to an Arduino Breadboard and obtain signal outputs on our laptop. After various trial and error, as well as assistance from Xiao, we were able to setup a circuit design that displayed signal outputs of our MEG Device. These advancements to our project had been attempted in hopes of using the Compression Shirt MEG for Bluetooth respiration monitoring; however, our bluetooth chip was defective and left this improvement as a future direction for our project. Ultimately, we were only successful in connecting the MEG device to an Arduino Breadboard which enabled us to monitor signal outputs from our devices, yet this was a significant advancement as we did not have to rely solely on the Preamplifier Device for testing.

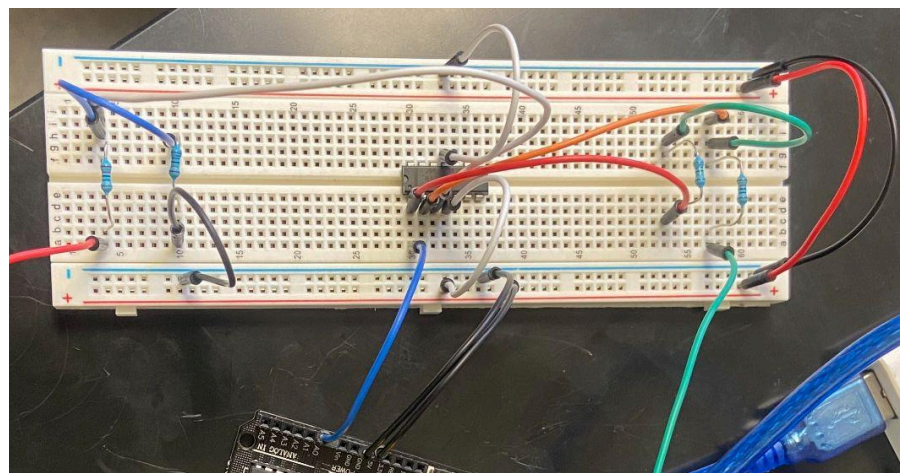
### **Test Day #3 (Assembling Arduino Components)**



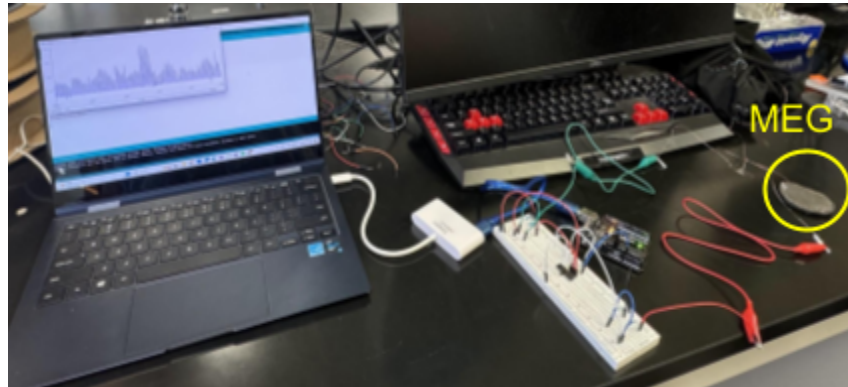
**Figure 7. Differential Amplifier Design**



**Figure 8. Group Members working on Arduino Breadboard**

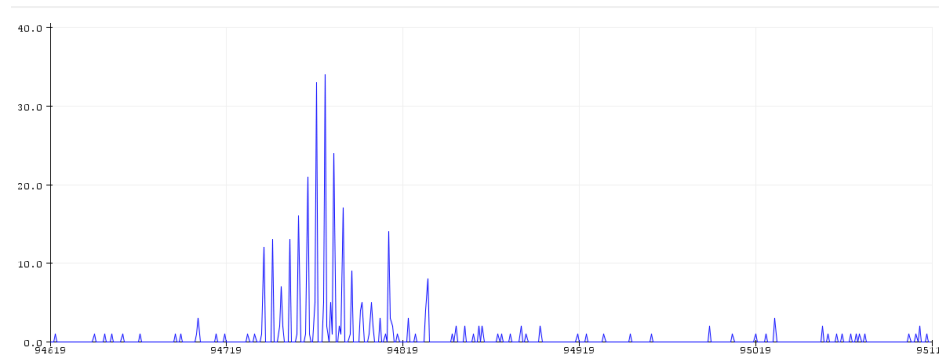


**Figure 9. Finished Arduino Circuit Design**

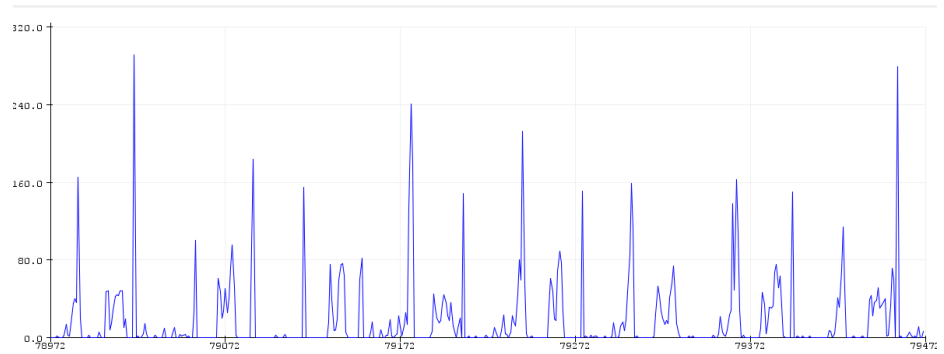


**Figure 10. Full Testing Setup with Arduino Components**

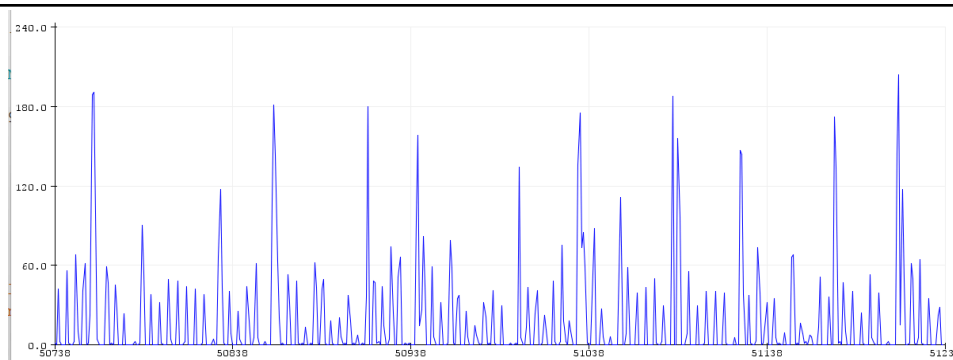
The copper wires of the MEG device are connected to the red and green leads which are then connected to the Arduino Breadboard/Microcontroller



**Graph 13. MEG Device Monitoring Slow Breathing Patterns**



**Graph 13. MEG Device Monitoring Rapid Breathing Patterns**



**Graph 13. MEG Device Monitoring Deep Breathing Patterns**

### **Test Day #3 (Arduino Components Analysis)**

Because we were integrating our MEG sensor with an Arduino, we needed a voltage output from the MEG of at least 0.25 V as that is the Arduino voltage input threshold. We designed a differential amplifier to help us reach this threshold voltage as we needed a gain in voltage, and because the MEG sensor gives relatively small voltage outputs with sensitive movements such as breathing we used 10K Ohm resistors for R1 and 1M Ohm resistors for R2 giving us a gain of 100 (From Figure.7). This ended up working great and with the proper program and code, we were able to read the voltage outputs from our MEG on the computer.

### **Conclusion**

On test day one, the MEG 1.0 pocket which housed the copper coils was created with a 1:1 ratio of Ecoflex to magnetic powder. Additionally, three variations of copper coil turns (25, 50, and 100) were tested to see which coil produced the highest current output. While 50 and 100 coil turns had similar current output, the 100 coil turns had the least amount of noise and was the most stable. However, the current output was not as strong as desired and could not detect slower breathing.

On test day two, after input from the TA, larger coil turns that filled the MEG pocket was tested. The larger coils nearly doubled the current output, improving detection of all breathing types. The MEG was tested on multiple different team members to see different breathing patterns. Each team member produced a different signal to noise ratio and a different amount of current, highlighting that data analysis needs to be individualized for each user.

On test day three, the MEG 2.0 was created which had a 1:2 Ecoflex to magnetic powder ratio. The increase in magnetic nanoparticles further increased the current output, allowing for sensitive enough detection for deep breathing. Additionally, an arduino circuit was integrated with the MEG to obtain a signal directly to a laptop, which is more portable than a lab ammeter. While we attempted to integrate a bluetooth chip, the chip could not connect to any laptop which is a future application we hope to work on.

The final MEG product showed its capability in detecting different breathing types, making it a useful device for athletes and health monitoring. Furthermore, MEGs are water resistant and have high current outputs, making the MEG well suited for active motions and small pressure sensing. MEGs revolutionized the healthcare field by offering a non-toxic, biocompatible and durable energy alternative to batteries. With the MEG compression shirt, an alternative and improved method of respiration monitoring is created which can serve a variety of healthcare needs.

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| <b>References</b> | Chen, J. Lecture Slides (2024) |